University

Management System:

Teamwork:

Student Name	Student Id
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Course: CESD215 – Introduction to Database

Section: T (22021)

Narrative Description:

The University Management System aims to develop a centralized database to manage information related to students, instructors, courses, departments, and academic projects.

Universities often face challenges when handling student records, project tracking, and course management using manual systems. This database will provide an efficient, accurate, and reliable platform to store and retrieve all relevant academic and administrative data, reducing errors and improving accessibility.

Identification of Information Needs:

The system requires accurate and organized data to manage university operations effectively. The main information needed includes:

- Student's personal and academic details.
- Instructor and professor information.
- Courses and their related departments.
- Student enrollments and grades. • Project information for students and professors.

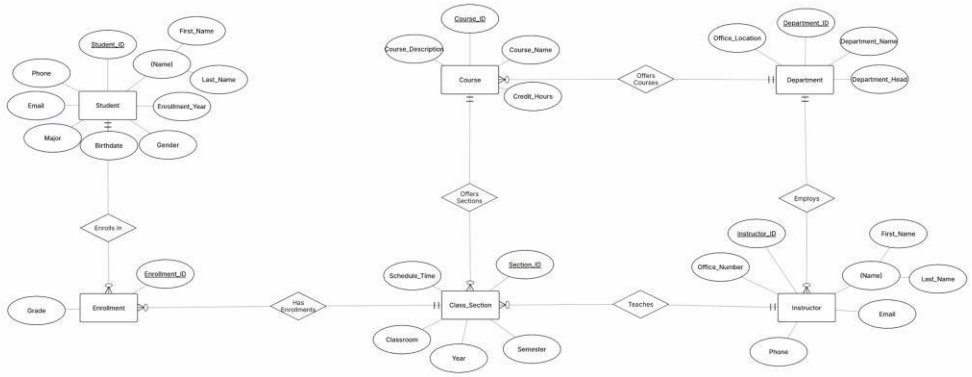
This data will help in simplifying academic management and improving communication between students and faculty.

Initial List of Entities (Tables)

The initial entities identified for the system are:

- **Student** (Student_ID, Name, Email, Major, Department_ID)
- **Professor** (Professor_ID, Name, Email, Department_ID)
- **Instructor** (Instructor_ID, Name, Email, Department_ID)
- **Department** (Department_ID, Department_Name, Location)
- **Course** (Course_ID, Course_Name, Credit_Hours, Department_ID)
- **Enrollment** (Enrollment_ID, Student_ID, Course_ID, Grade)
- **Project** (Project_ID, Project_Title, Description, Professor_ID, Student_ID, Grade)

The first step ERD:



The second step, Mapping:

STUDENTS

Students_ID	First_Name	Last_Name	Email	Major	Phone	Birthdate	Gender	Enrolment_Year

INSTRUCTOR

Instructor_ID	First_Name	Last_name	Email	Phone	Office_Number	~~Department_ID~~

DEPARTMENT

Department_ID	Department_Name	~~Department_Head~~	Office_location

COURSE

Course_ID	Course_Name	Course_Description	Credit_Hours	~~Department_ID~~

CLASS_SECTION

Section_ID	~~Course_ID~~	~~Instructor_ID~~	Semester	Schedule_Time	Classroom	Year

ENROLLMENT

Enrollment_ID	Student_ID	Section_ID	Grade

Functional Dependencies:

TABEL	FD
STUDENTS	Student_ID (PK) → First_Name, Last_Name, Email, Major, Phone, Birthdate, Gender, Enrollment_Year
INSTRUCTOR	Instructor_ID(PK) → First_Name, Last_Name, Email, Phone, Office_Number, Department_ID(FK)
DEPARTMENT	Department_ID (PK) → Department_Name, Department_Head(FK), Office_location
COURSE	Course_ID(PK) → Course_Name, Course_Description, Credit_Hours, Department_ID(FK)
CLASS_SECTION	Section_ID(PK) → Course_ID(FK), Instructor_ID(FK), Semester, Schedule_Time, Classroom, Year
ENROLLMENT	Enrollment_ID (PK)→ Student_ID(FK), Section_ID(FK), Grade

Normalization

(1NF → 2NF → 3NF)

1. **STUDENTS**

Primary Key: Student_ID

FD: Student_ID → all other attributes

1NF

- All attributes contain atomic values.
- No repeating groups.

→1NF satisfied

2NF

- The primary key is a single attribute (Student_ID).Therefore, no partial dependencies can exist.

→ 2NF satisfied

3NF

- No non-key attribute depends on another non-key attribute.
- Every attribute depends directly on the key.

→ 3NF satisfied

2.

INSTRUCTOR

Primary Key: Instructor_ID

FD: Instructor_ID → all other attributes

1NF

- All fields are atomic.

→ 1NF satisfied

2NF

- Key is single, so partial dependency is impossible.

→ 2NF satisfied

3NF

- Non-key attributes do not determine other non-key attributes.
- Department_ID is a foreign key but does not violate 3NF.

→ 3NF satisfied

3.

DEPARTMENT

Primary Key: Department_ID

FD: Department_ID → Department_Name, Department_Head, Office_Location

1NF

- All fields contain atomic values.

→ 1NF satisfied

2NF

- Single primary key → no partial dependency.

→ 2NF satisfied

3NF

- No transitive dependency between non-key attributes.

→ 3NF satisfied

4.

COURSE

Primary Key: Course_ID

FD: Course_ID → Course_Name, Course_Description, Credit_Hours, Department_ID

1NF

- Attributes are atomic.

→ 1NF satisfied

2NF

- Single primary key so no partial dependency

→ 2NF satisfied

3NF

- Non-key attributes do not depend on each other.
- Department_ID is a foreign key only.

→ 3NF satisfied

5.

CLASS_SECTION

Primary Key: Section_ID

FD: Section_ID → Course_ID, Instructor_ID, Semester, Schedule_Time, Classroom, Year

1NF

- All values are atomic; no repeating groups.

→ 1NF satisfied

2NF

- Single-column primary key → no partial dependency.

→ 2NF satisfied

3NF

- No non-key attribute determines another non-key attribute.

→ 3NF satisfied

6.

ENROLLMENT

Primary Key: Enrollment_ID

FD: Enrollment_ID → Student_ID, Section_ID, Grade

1NF

- All fields are atomic

→ 1NF satisfied

2NF

- The Primary Key it is a single attribute → no partial dependency.

→ 2NF satisfied

3NF

- No transitive dependencies among non-key attributes.

→ 3NF satisfied

Table	Attribute	Datatype	Constraints
Students	Student_ID	NUMBER	PK
	First_Name	VARCHAR2(50)	Not Null
	Last_Name	VARCHAR2(50)	Not Null
	Email	VARCHAR2(100)	Unique
	Major	VARCHAR2(100)	
	Phone	VARCHAR2(20)	Uniqu
	Birthdate	DATE	
	Gender	VARCHAR2(10)	Check (Male/Female)
	Enrolment_Year	NUMBER	
Instructor	Instructor_ID	NUMBER	PK
	First_Name	VARCHAR2(50)	Not Null
	Last_Name	VARCHAR2(50)	Not Null
	Email	VARCHAR2(100)	Unique
	Phone	VARCHAR2(20)	Unique
	Office_Number	VARCHAR2(20)	
	Department_ID	NUMBER	FK, Not Null
Department	Department_ID	NUMBER	PK
	Department_Name	VARCHAR2(100)	Unique
	Department_Head	VARCHAR2(100)	
	Office_Location	VARCHAR2(100)	
Course	Course_ID	NUMBER	PK
	Course_Name	VARCHAR2(100)	Not Null
	Course_Description	VARCHAR2(200)	
	Credit_Hours	NUMBER	Check (Credit_Hours > 0)
	Department_ID	NUMBER	FK, Not Null
Class_Section	Section_ID	NUMBER	PK
	Course_ID	NUMBER	FK, Not Null
	Instructor_ID	NUMBER	FK, Not Null
	Semester	VARCHAR2(20)	Check ('Fall','Spring','Summer')
	Schedule_Time	VARCHAR2(50)	
	Classroom	VARCHAR2(20)	
	Year	NUMBER	Check (Year >= 2000)
Enrollment	Enrollment_ID	NUMBER	PK
	Student_ID	NUMBER	FK, Not Null
	Section_ID	NUMBER	FK, Not Null

	Grade	VARCHAR2(5)	Check (Grade IN ('A','B','C','D','F','W'))

Creating Tables

1.Students' Table

```sql
CREATE TABLE Students (
    Student_ID NUMBER PRIMARY KEY,
    First_Name VARCHAR2(50) NOT NULL,
    Last_Name VARCHAR2(50) NOT NULL,
    Email VARCHAR2(100) UNIQUE,
    Major  VARCHAR2(100),
    Phone  VARCHAR2(20) UNIQUE,
    Birthdate DATE,
    Gender VARCHAR2(10)
        CONSTRAINT ck_students_gender CHECK (Gender IN ('Male', 'Female')),
    Enrolment_Year NUMBER
);
```

```
Table created.
```

2.Instructor Table

```sql
CREATE TABLE Instructor (
    Instructor_ID NUMBER PRIMARY KEY,
    First_Name VARCHAR2(50) NOT NULL,
    Last_Name VARCHAR2(50) NOT NULL,
    Email  VARCHAR2(100) UNIQUE,
    Phone VARCHAR2(20) UNIQUE,
    Office_Number VARCHAR2(20),
    Department_ID  NUMBER NOT NULL
);
```

```
Table created.
```

3.Department Table

```
CREATE TABLE Department (
    Department_ID NUMBER PRIMARY KEY,
    Department_Name VARCHAR2(100) UNIQUE,
    Department_Head VARCHAR2(100),
    Office_Location VARCHAR2(100)
);
```

```
Table created.
```

4.Course Table

```
CREATE TABLE Course (
    Course_ID NUMBER PRIMARY KEY,
    Course_Name  VARCHAR2(100) NOT NULL,
    Course_Description VARCHAR2(200),
    Credit_Hours NUMBER
        CONSTRAINT ck_credit_hours CHECK (Credit_Hours > 0),
    Department_ID NUMBER NOT NULL
);
```

```
Table created.
```

5.Class_Section Table

```sql
CREATE TABLE Class_Section (
    Section_ID NUMBER PRIMARY KEY,
    Course_ID NUMBER NOT NULL,
    Instructor_ID NUMBER NOT NULL,
    Semester  VARCHAR2(20)
        CONSTRAINT ck_section_semester CHECK (Semester IN ('Fall','Spring','Summer')),
    Schedule_Time VARCHAR2(50),
    Classroom VARCHAR2(20),
    Year NUMBER
        CONSTRAINT ck_section_year CHECK (Year >= 2000)
);
```

```
Table created.
```

6.Enrollment Table

```sql
CREATE TABLE Enrollment (
    Enrollment_ID NUMBER PRIMARY KEY,
    Student_ID NUMBER NOT NULL,
    Section_ID  NUMBER NOT NULL,
    Grade  VARCHAR2(5)
        CONSTRAINT ck_enroll_grade CHECK (Grade IN ('A','B','C','D','F','W'))
);
```

```
Table created.
```

ALTER TABLE

1. Instructor → Department

```sql
ALTER TABLE Instructor
ADD CONSTRAINT fk_instructor_dept
FOREIGN KEY (Department_ID)
REFERENCES Department(Department_ID);
```

```
Table altered.
```

2. Course → Department

```sql
ALTER TABLE Course
ADD CONSTRAINT fk_course_dept
FOREIGN KEY (Department_ID)
REFERENCES Department(Department_ID);
```

```
Table altered.
```

3. Class_Section → Course

```sql
ALTER TABLE Class_Section
ADD CONSTRAINT fk_section_course
FOREIGN KEY (Course_ID)
REFERENCES Course(Course_ID);
```

```
Table altered.
```

4. Class_Section → Instructor

```sql
ALTER TABLE Class_Section
ADD CONSTRAINT fk_section_instructor
FOREIGN KEY (Instructor_ID)
REFERENCES Instructor(Instructor_ID);
```

```
Table altered.
```

5. Enrollment → Students

```sql
ALTER TABLE Enrollment
ADD CONSTRAINT fk_enroll_student
FOREIGN KEY (Student_ID)
REFERENCES Students(Student_ID);
```

```
Table altered.
```

6. Enrollment → Class_Section

```sql
ALTER TABLE Enrollment
ADD CONSTRAINT fk_enroll_section
FOREIGN KEY (Section_ID)
REFERENCES Class_Section(Section_ID);
```

Table altered.

Students table

Insert

```sql
INSERT INTO Students VALUES
(1, 'Sara', 'Ali', 'sara.ali@gmail.com', 'Computer Science', '0551000001', DATE '2004-05-10', 'Female', 2022);
INSERT INTO Students VALUES
(2, 'Omar', 'Hassan', 'omar.hassan@gmail.com', 'Information Systems', '0551000002', DATE '2003-11-20', 'Male', 2021);
INSERT INTO Students VALUES
(3, 'Lina', 'Saad', 'lina.saad@gmail.com', 'Cyber Security', '0551000003', DATE '2005-01-08', 'Female', 2023);
INSERT INTO Students VALUES
(4, 'Fahad', 'Nasser', 'fahad.nasser@gmail.com', 'Software Engineering', '0551000004', DATE '2002-09-15', 'Male', 2020);
INSERT INTO Students VALUES
(5, 'Mona', 'Yousef', 'mona.yousef@gmail.com', 'Data Science', '0551000005', DATE '2004-03-18', 'Female', 2022);
SELECT * FROM Students;
```

Select

STUDENT_ID	FIRST_NAME	LAST_NAME	EMAIL	MAJOR	PHONE	BIRTHDATE	GENDER	ENROLMENT_YEAR
1	Sara	Ali	sara.ali@gmail.com	Computer Science	0551000001	10-MAY-04	Female	2022
2	Omar	Hassan	omar.hassan@gmail.com	Information Systems	0551000002	20-NOV-03	Male	2021
3	Lina	Saad	lina.saad@gmail.com	Cyber Security	0551000003	08-JAN-05	Female	2023
4	Fahad	Nasser	fahad.nasser@gmail.com	Software Engineering	0551000004	15-SEP-02	Male	2020
5	Mona	Yousef	mona.yousef@gmail.com	Data Science	0551000005	18-MAR-04	Female	2022

Download CSV

5 rows selected.

Instructor Table

Insert

```sql
INSERT INTO Instructor VALUES
(10, 'Khalid', 'Saleh', 'khalid.saleh@gmail.com', '0552000001', 'B203', 1);
INSERT INTO Instructor VALUES
(11, 'Amani', 'Yousef', 'amani.yousef@gmail.com', '0552000002', 'C109', 2);
INSERT INTO Instructor VALUES
(12, 'Huda', 'Othman', 'huda.othman@gmail.com', '0552000003', 'A310', 1);
INSERT INTO Instructor VALUES
(13, 'Saeed', 'Mahdi', 'saeed.mahdi@gmail.com', '0552000004', 'B120', 3);
INSERT INTO Instructor VALUES
(14, 'Nora', 'Faleh', 'nora.faleh@gmail.com', '0552000005', 'C220', 2);

SELECT * FROM Instructor;
```

Select

INSTRUCTOR_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE	OFFICE_NUMBER	DEPARTMENT_ID
10	Khalid	Saleh	khalid.saleh@gmail.com	0552000001	B203	1
11	Amani	Yousef	amani.yousef@gmail.com	0552000002	C109	2
12	Huda	Othman	huda.othman@gmail.com	0552000003	A310	1
13	Saeed	Mahdi	saeed.mahdi@gmail.com	0552000004	B120	3
14	Nora	Faleh	nora.faleh@gmail.com	0552000005	C220	2

Download CSV

5 rows selected.

Department Table

Insert

```sql
INSERT INTO Department VALUES
(1, 'Computer Science', 'Dr. Khalid Saleh', 'Building B');
INSERT INTO Department VALUES
(2, 'Information Systems', 'Dr. Amani Yousef', 'Building C');
INSERT INTO Department VALUES
(3, 'Cyber Security', 'Dr. Saeed Mahdi', 'Building A');
INSERT INTO Department VALUES
(4, 'Data Science', 'Dr. Lina Omar', 'Building D');
INSERT INTO Department VALUES
(5, 'Software Engineering', 'Dr. Fahad Ali', 'Building E');

SELECT * FROM Department;
```

Select

DEPARTMENT_ID	DEPARTMENT_NAME	DEPARTMENT_HEAD	OFFICE_LOCATION
1	Computer Science	Dr. Khalid Saleh	Building B
2	Information Systems	Dr. Amani Yousef	Building C
3	Cyber Security	Dr. Saeed Mahdi	Building A
4	Data Science	Dr. Lina Omar	Building D
5	Software Engineering	Dr. Fahad Ali	Building E

Download CSV

5 rows selected.

Course Table

Insert

```sql
INSERT INTO Course VALUES
(101, 'Database Systems', 'Intro to databases', 3, 1);
INSERT INTO Course VALUES
(102, 'Programming I', 'Introduction to programming', 4, 1);
INSERT INTO Course VALUES
(103, 'Network Security', 'Principles of security', 3, 3);
INSERT INTO Course VALUES
(104, 'Data Mining', 'Basic data analysis', 3, 4);
INSERT INTO Course VALUES
(105, 'Software Design', 'Software design principles', 3, 5);
SELECT * FROM Course;
```

Select

COURSE_ID	COURSE_NAME	COURSE_DESCRIPTION	CREDIT_HOURS	DEPARTMENT_ID
101	Database Systems	Intro to databases	3	1
102	Programming I	Introduction to programming	4	1
103	Network Security	Principles of security	3	3
104	Data Mining	Basic data analysis	3	4
105	Software Design	Software design principles	3	5

Download CSV

5 rows selected.

Class_Section Table

Insert

```sql
INSERT INTO Class_Section VALUES
(501, 101, 10, 'Fall', 'Mon 10-12', 'B12', 2024);
INSERT INTO Class_Sectio VALUES
(502, 102, 11, 'Spring', 'Wed 1-3', 'C07', 2024);
INSERT INTO Class_Section VALUES
(503, 103, 13, 'Fall', 'Sun 9-11', 'A15', 2024);
INSERT INTO Class_Section VALUES
(504, 104, 12, 'Summer', 'Tue 12-2', 'D09', 2023);
INSERT INTO Class_Section VALUES
(505, 105, 14, 'Spring', 'Thu 2-4', 'E03', 2024);

SELECT * FROM Class_Section;
```

Select

SECTION_ID	COURSE_ID	INSTRUCTOR_ID	SEMESTER	SCHEDULE_TIME	CLASSROOM	YEAR
501	101	10	Fall	Mon 10-12	B12	2024
503	103	13	Fall	Sun 9-11	A15	2024
504	104	12	Summer	Tue 12-2	D09	2023
505	105	14	Spring	Thu 2-4	E03	2024

Download CSV

4 rows selected.

Enrolment Table

Insert

```sql
INSERT INTO Enrollment VALUES
(9001, 1, 501, 'A');
INSERT INTO Enrollment VALUES
(9002, 2, 502, 'B');
INSERT INTO Enrollment VALUES
(9003, 3, 503, 'C');
INSERT INTO Enrollment VALUES
(9004, 4, 504, 'A');
INSERT INTO Enrollment VALUES
(9005, 5, 505, 'B');

SELECT * FROM Enrollment;
```

Select

ENROLLMENT_ID	STUDENT_ID	SECTION_ID	GRADE
9001	1	501	A
9002	2	502	B
9003	3	503	C
9004	4	504	A
9005	5	505	B

Download CSV

5 rows selected.

Queries:

Query using GROUP BY + COUNT:

```sql
SELECT
    Major,
    COUNT(*) AS Number_Of_Students
FROM Students
GROUP BY Major;
```

MAJOR	NUMBER_OF_STUDENTS
Data Science	1
Information Systems	1
Cyber Security	1
Software Engineering	1
Computer Science	1

Download CSV

5 rows selected.

Query using GROUP BY + AVG:

```sql
SELECT
    Department_ID,
    AVG(Credit_Hours) AS Avg_Credit_Hours
FROM Course
GROUP BY Department_ID;
```

DEPARTMENT_ID	AVG_CREDIT_HOURS
1	3.5
4	3
5	3
3	3

Download CSV

4 rows selected.

Query using Subquery + Aggregate Function:

```sql
SELECT
    Course_ID,
    Course_Name,
    Credit_Hours
FROM Course
WHERE Credit_Hours > (
    SELECT AVG(Credit_Hours)
    FROM Course
);
```

COURSE_ID	COURSE_NAME	CREDIT_HOURS
102	Programming I	4

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Query using Subquery (IN):

```sql
SELECT
    Student_ID,
    First_Name,
    Last_Name
FROM Students
WHERE Student_ID IN (
    SELECT Student_ID
    FROM Enrollment
);
```

STUDENT_ID	FIRST_NAME	LAST_NAME
1	Sara	Ali
2	Omar	Hassan
3	Lina	Saad
4	Badr	Mansour
5	Mona	Yousef

Download CSV

5 rows selected.

Query using Multiple JOINs :

```sql
-- Students with their enrolled courses
SELECT s.First_Name, s.Last_Name, c.Course_Name
FROM Students s
JOIN Enrollment e ON s.Student_ID = e.Student_ID
JOIN Class_Section cs ON e.Section_ID = cs.Section_ID
JOIN Course c ON cs.Course_ID = c.Course_ID;
```

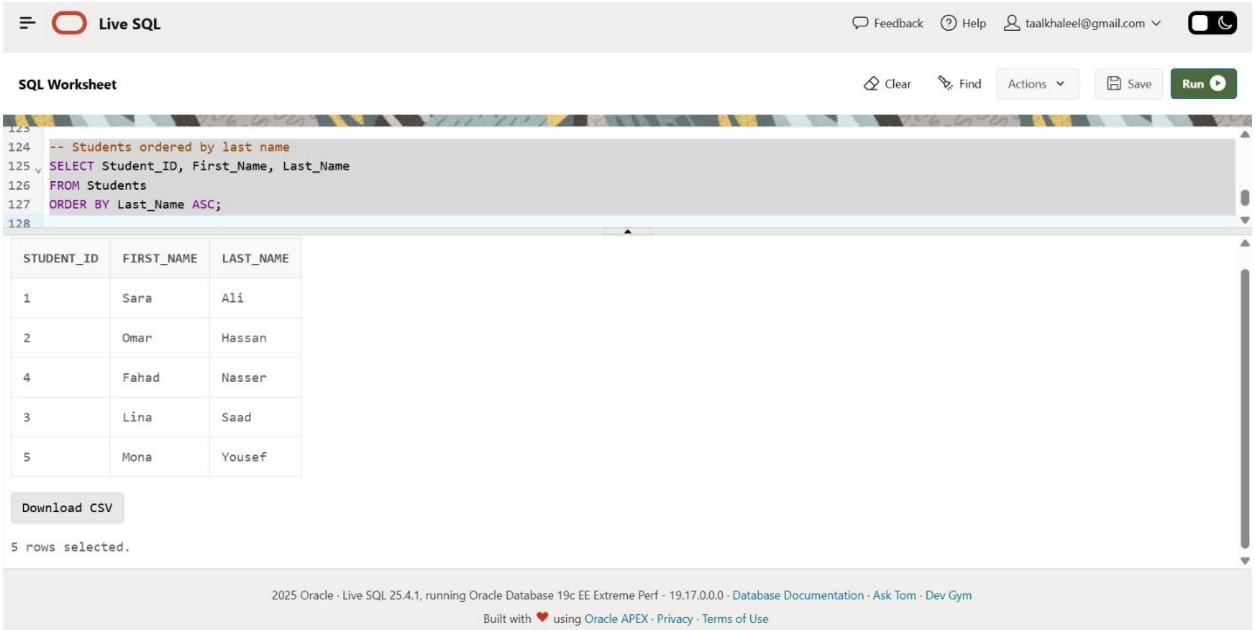
SQL Worksheet Clear Find Actions Save Run

FIRST_NAME	LAST_NAME	COURSE_NAME
Sara	Ali	Database Systems
Omar	Hassan	Programming I
Lina	Saad	Network Security
Fahad	Nasser	Data Mining
Mona	Yousef	Software Design

Download CSV

5 rows selected.

Query using ordered by last name:



Query using WHERE :

SQL Worksheet

Clear Find Actions Save Run

```sql
SELECT Course_ID, Course_Name, Credit_Hours
FROM Course
WHERE Credit_Hours BETWEEN 3 AND 5;
```

COURSE_ID	COURSE_NAME	CREDIT_HOURS
101	Database Systems	3
102	Programming I	4
103	Network Security	3
104	Data Mining	3
105	Software Design	3

Download CSV

5 rows selected.